HOMEWORK ASSIGNMENT 7

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

QUESTION 1: Picture It!

In Quiz 05, we briefly explored a 2025[1] study, designed to explore the data visualization course offerings across several highly-ranked (as determined by US News) US higher-education institutes. In this problem, we'll explore the data a bit further. In particular, 50 of the 154 institutes included in the sample are classified as "liberal arts colleges" whereas the remaining 104 are classified as "universities." Among the liberal arts colleges, 16 offered at least one data visualization course and the remaining 34 did not. Among the universities, 78 offered at least one data visualization course and the remaining 26 did not.

(a) Fill in the following two-way table with the appropriate values.

	Liberal Arts College	**University**	**Total**
At Least One Data Viz. Course Offered			
No Data Viz. Course Offered			
Total			

(b) Calculate the conditional proportion of liberal arts colleges that offer at least one data visualization course, and interpret this value.

(c) Sketch a segmented bar graph of the data contained in the two-way table.

(d) Sketch a mosaic plot of the data contained in the two-way table.

(e) Calculate the difference in conditional proportions of data visualization course offerings between liberal arts colleges and universities.

(f) Calculate the relative risk, ensuring that your final answer is greater than 1, and interpret this value.

(g) Calculate and interpret the odds ratio comparing the odds of offering at least one data visualization course for the liberal arts colleges to the odds of offering at least one data visualization course for the universities. Interpret this value.

[1] Branson, Z., Paz Parra, M., & Yurko, R. (2025). The Landscape of College-Level Data Visualization Courses, and the Benefits of Incorporating Statistical Thinking. *Journal of Statistics and Data Science Education*, 33(4), 390–406. https://doi.org/10.1080/26939169.2025.2537049

QUESTION 2: Picture It! (Continued)

Consider again the setup of Question 1 above, along with the data described in it. In this question, we'll explore the following research question: are highly-ranked US universities more likely to offer at least one data visualization course than highly-ranked US liberal arts colleges? **Note:** below, we'll use "Univ." as an abbreviation for "University," and "LAC" as an abbreviation for "Liberal Arts College." Additionally, wherever necessary, use the order (University − LAC) and treat "success" as "offering at least one data visualization course."

(a) Calculate the value of the statistic (use the *difference* of sample proportions, and use University − Liberal Arts College).

(b) What are the appropriate symbols to represent the statistic?

- ◯ $\pi_{\text{Univ}} - \pi_{\text{LAC}}$
- ◯ $\hat{p}_{\text{Univ}} - \hat{p}_{\text{LAC}}$
- ◯ $\hat{p}_{\text{Univ}-\text{LAC}}$
- ◯ $\pi_{\text{Univ}-\text{LAC}}$

(c) Write down the null and alternative hypotheses using symbols and math only, but also provide a verbal description of any and all symbols you use.

(d) Are the validity conditions met to conduct a theory-based test to answer the research question? Do not just state the validity conditions. Use the data from the sample to explain how the validity conditions are or are not met.

(e) Use the Theory-based Inference applet to carry out a two-sample z-test to answer the research question.

(f) Based on the *p*-value you found, do you reject or fail to reject the null hypothesis?

- ◯ Reject the null hypothesis in favor of the alternative hypothesis
- ◯ Fail to reject the null hypothesis

(g) Given the results of your analysis, provide an overall conclusion **in context**. Be very clear what you can claim about the population proportions. IF you have evidence that one population proportion is greater/smaller/different than the other, then you need to state which one is greater/smaller/different.

(h) To which population can you generalize the results of your study?

- ◯ We could generalize the results to all higher education institutes in the US
- ◯ We could generalize the results to higher-education institutes in the US that are currently (in 2026) highly ranked by US News
- ◯ We could generalize the results to higher-education institutes in the US that were highly ranked by US News in 2025

(i) Assuming we found a statistically significant result (which isn't necessarily the right answer to any of the previous parts!), could we make a **causal** claim? For instance, could we say that the classification of a higher-education institute (as a University or a Liberal Arts College) was the *cause* for offering or not offering a data visualization course, in 2025? Explain.

(j) Without using any applets, calculate a 90% confidence interval for the difference in the population proportions of Universities and Liberal Arts Colleges that offer at least one data visualization course.

(k) Use the Theory-Based Inference applet to calculate a 90% confidence interval for the difference in the population proportions of Universities and Liberal Arts Colleges that offer at least one data visualization course.

(l) Interpret the interval you constructed in Part (k) above, using the **context** of the problem. Be sure to use words such as **greater** and/or **smaller** in your interpretation to make it clear if there is a population proportion higher than the other.

QUESTION 3: What Are the Odds?

Imagine we conducted a randomized experiment in which subjects were administered one of two doses ("A" or "B") of a medicine, and observed to see whether or not the subject developed an adverse reaction ("yes" or "no"). The following two-way table was created from the data.

	Dose A	Dose B	Total
Adverse Reaction	20	40	**60**
No Adverse Reaction	30	50	**80**
Total	**50**	**90**	**140**

(a) Calculate the conditional proportion of subjects on Dose A that developed an adverse reaction.

(b) Compute the relative risk of developing adverse reactions between the two doses, and interpret this value. **Hint:** remember that it's recommended to set up the relative risk to be greater than or equal to 1.

(c) Calculate the odds of not developing adverse reactions while on Dose A, and interpret this value.

(d) Calculate and interpret the odds ratio comparing the odds of developing adverse reactions under Dose A to the odds of developing adverse reactions under Dose B. Interpret this value.

(e) **Without doing any additional calculations**: Consider the quantity "the odds ratio comparing the odds of not developing adverse reactions under Dose B to the odds of not developing adverse reactions under Dose A." How would this quantity compare to the value you calculated in Part (d) above

- ◯ It would be the same
- ◯ The quantity in this part would be the reciprocal of the quantity in Part (d) above

(f) **Without doing any additional calculations**: Consider the quantity "the odds ratio comparing the odds of not developing adverse reactions under Dose A to the odds of not developing adverse reactions under Dose B." How would this quantity compare to the value you calculated in Part (d) above?

- ◯ It would be the same
- ◯ The quantity in this part would be the reciprocal of the quantity in Part (d) above.